

fong-spivak-invitation: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_E08282 [conf 0.001]	291	■ 0.001	$G_1 = \begin{array}{ c } \hline \text{ } \bullet \xrightarrow{f} \bullet \\ \hline f = g \end{array}$ $G_2 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline f = a \end{array}$ $G_3 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \begin{array}{c} g \downarrow \bullet \xrightarrow{i} \bullet \\ \downarrow h \end{array} \\ f \circ h = g \circ i \end{array}$ $G_4 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \begin{array}{c} g \downarrow \bullet \xrightarrow{h} \bullet \\ \downarrow h \end{array} \\ \text{no equations} \end{array}$	(not rendered)	
fong-spivak-invitation_E0862 [conf 0.001]	113	■ 0.001	$I = \left(\begin{array}{ccc} 1 & a & 2 \\ \downarrow & & \downarrow \\ \triangleright & 3 & \\ & 4 & \\ & b & \end{array} \right)$		$I = \begin{array}{ c } \hline \begin{array}{ccc} 1 & \xrightarrow{a} & 2 \\ & \searrow b & \downarrow e \\ & & 3 \end{array} \\ \hline \end{array}$
fong-spivak-invitation_E08180 [conf 0.013]	273	■ 0.013	$G = \left(\begin{array}{ccc} 1 & & 2 \\ & a \downarrow e \downarrow & \\ & c & v_3 \end{array} \right)$		$G = \begin{array}{ c } \hline \begin{array}{ccc} 1 & \xrightarrow{a} & 2 \\ & \searrow b & \downarrow e \\ & & 3 \end{array} \\ \hline \end{array}$
fong-spivak-invitation_E08135 [conf 0.019]	215	■ 0.019	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\text{id}_X, \text{id}_X} X \xleftarrow{\text{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\text{id}_X} X \xrightarrow{\text{id}_X, \text{id}_X} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\text{id}_X} \emptyset \right) \end{aligned}$	$\mu_X := (X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X)$ $\eta_X := (\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X)$ $\delta_X := (X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X)$ $\epsilon_X := (X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset)$	$\mu_X := (X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X)$ $\eta_X := (\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X)$ $\delta_X := (X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X)$ $\epsilon_X := (X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset)$

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fong-spivak-invitation_ EQ0023	58	■ 0.254	$\left.\overline{\frac{3.14}{\frac{3.14}{-1}}}\right\}=\frac{3.14}{-1}\quad \overline{0}=\text{nothing}$	$\overline{3.14}\quad \overline{-1}\quad \overline{\frac{3.14}{-1}}\Big\}=\frac{3.14}{-1}\quad \overline{0}=\text{nothing}$	$\overline{3.14}\quad \overline{-1}\quad \overline{\frac{3.14}{-1}}\Big\}=\frac{3.14}{-1}\quad \overline{0}=\text{nothing}$
fong-spivak-invitation_ EQ0227	306	■ 0.259	$\begin{aligned} &\bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\mathcal{X}(x,a)\otimes\mathcal{X}(a,b)\otimes\mathcal{X}(c,d)\otimes\mathcal{X}(b,c)\otimes\mathcal{X}(d,e)\otimes\mathcal{X}(e,y) \\ &= \mathcal{X}(x,y), \end{aligned}$	$\begin{aligned} &\bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\mathcal{X}(x,a)\otimes\epsilon_{\mathcal{X}}(b,c,1)\otimes\mathrm{U}_{\mathcal{X}}(d,e)\otimes\alpha((1,e),y) \\ &= x(x,y), \end{aligned}$	$\begin{aligned} &\bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\alpha^{-1}(x,(a,1))\otimes\mathrm{U}_{\mathcal{X}}(a,b)\otimes\eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e))\otimes\alpha((1,e),y) \\ &= \bigvee_{a,b,c,d,e\in\mathcal{X}}\mathcal{X}(x,a)\otimes\mathcal{X}(a,b)\otimes\mathcal{X}(c,d)\otimes\mathcal{X}(b,c)\otimes\mathcal{X}(d,e)\otimes\mathcal{X}(e,y) \\ &= \mathcal{X}(x,y), \end{aligned}$
fong-spivak-invitation_ EQ0040	78	■ 0.267	$\begin{array}{l} \mathcal{C}_{\mathcal{C}}(c,d)\otimes_W\mathcal{C}_{\mathcal{C}}(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}) \\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\quad (f\text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c,e))\quad (\mathcal{C}\text{ is }\mathcal{V}\text{-category}) \\ =\mathcal{C}_{\mathcal{C}}(c,e)\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}). \end{array}$	$\begin{array}{l} \mathcal{C}_{\mathcal{C}}(c,d)\otimes_W\mathcal{C}_{\mathcal{C}}(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}) \\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\quad (f\text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c,e))\quad (\mathcal{C}\text{ is }\mathcal{V}\text{-category}) \\ =\mathcal{C}_{\mathcal{C}}(c,e)\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}). \end{array}$	$\begin{array}{l} \mathcal{C}_{\mathcal{C}}(c,d)\otimes_W\mathcal{C}_{\mathcal{C}}(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}) \\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\quad (f\text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c,e))\quad (\mathcal{C}\text{ is }\mathcal{V}\text{-category}) \\ =\mathcal{C}_{\mathcal{C}}(c,e)\quad (\text{definition of }\mathcal{C}_{\mathcal{C}}). \end{array}$
fong-spivak-invitation_ EQ0256	320	■ 0.281	$\left[\mathrm{id}_A,!_A\right]^{\circ}\iota_A=\left[\mathrm{id}_A^{\circ}\iota_A,!_A^{\circ}\iota_A\right]=\left[\iota_A,!_{A+\varnothing}\right]=\left[\iota_A,\iota_{\varnothing}\right]=\mathrm{id}_{A+\varnothing}.$	$[\mathrm{id}_A,!_A]^{\circ}\iota_A=[\mathrm{id}_A^{\circ}\iota_A,!_A^{\circ}\iota_A]=[\iota_A,!_{A+\varnothing}]=[\iota_A,\iota_{\varnothing}]=\mathrm{id}_{A+\varnothing}.$	$[\mathrm{id}_A,!_A]^{\circ}\iota_A=[\mathrm{id}_A^{\circ}\iota_A,!_A^{\circ}\iota_A]=[\iota_A,!_{A+\varnothing}]=[\iota_A,\iota_{\varnothing}]=\mathrm{id}_{A+\varnothing}.$
fong-spivak-invitation_ EQ0043 (2.22)	80	■ 0.305	$(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_2,y_2))\otimes(\mathcal{X}\times\mathcal{Y})((x_2,y_2),(x_3,y_3))\leq(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_3,y_3)).$	$(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_2,y_2))\otimes(\mathcal{X}\times\mathcal{Y})((x_2,y_2),(x_3,y_3))\leq(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_3,y_3)).$	$(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_2,y_2))\otimes(\mathcal{X}\times\mathcal{Y})((x_2,y_2),(x_3,y_3))\leq(\mathcal{X}\times\mathcal{Y})((x_1,y_1),(x_3,y_3)).$
fong-spivak-invitation_ EQ0112 (5.6)	177	■ 0.315	$G_{\mathcal{R}}:=\{\bigcirc-,\circ-,\subset,\cdots\}\cup\{-a-\mid a\in R\},$	$G_R:=\{\bigcirc-,\circ-,\subset,\cdots\}\cup\{-a-\mid a\in R\},$	$G_R:=\left\{\bigcirc-,\circ-,\subset,\cdots\right\}\cup\left\{-a-\mid a\in R\right\},$
fong-spivak-invitation_ EQ0059 (3.8)	108	■ 0.323	$\mathcal{C}:=\begin{array}{c} s \\ Q \\ \bullet \\ z \\ s;s=s \end{array}$	$\mathcal{C}:=\begin{array}{c} s \\ Q \\ \bullet \\ z \\ s;s=s \end{array}$	$\mathcal{C}:=\begin{array}{c} s \\ Q \\ \bullet \\ z \\ s;s=s \end{array}$

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fong-spivak-invitation_EQ0118	184	■ 0.327	$G_{\{R\}}:=\{\supset-, \circ-, \subset, \cdots\} \cup \{-\mathrm{a}- \mid \mathrm{a} \in R\},$	$G_R := \{\supset -, \circ -, \subset, \cdots\} \cup \{-\mathrm{a}- \mid \mathrm{a} \in R\},$	$G_R := \left\{ \supset -, \circ -, \subset -, \bullet - \right\} \cup \left\{ \neg \Box \mathrm{a} \mid \mathrm{a} \in R \right\},$
fong-spivak-invitation_EQ0037	74	■ 0.341	$X:=\operatorname{Ob}\{0b\}(\operatorname{\mathcal{X}}), \quad \quad d(x,y):=\operatorname{\mathcal{X}}(x,y) \ .$	$X := \operatorname{Ob}(\mathcal{X}), \quad d(x,y) := \mathcal{X}(x,y).$	$X := \operatorname{Ob}(\mathcal{X}), \quad d(x,y) := \mathcal{X}(x,y).$
fong-spivak-invitation_EQ0260	332	■ 0.372	$\bigcup_{i \in I} A_{\{i\}}=\bigcup_{i \in I} \left(B_{\{i\}} \cap Y \right)=\left(\bigcup_{i \in i} B_{\{i\}} \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in i} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in i} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$
fong-spivak-invitation_EQ0075 (3.15)	121	■ 0.380	$\operatorname{\mathcal{G}}:=\mid \stackrel{\text{Email}}{\text{received_by}} \stackrel{\text{sent_by}}{\text{Address}} \longmapsto \text{Address}$	<i>(not rendered)</i>	$\mathcal{G} := \begin{array}{c} \text{Email} \xrightarrow{\text{sent_by}} \text{Address} \\ \text{received_by} \\ \text{no equations} \end{array}$
fong-spivak-invitation_EQ0141	221	■ 0.382	$\begin{aligned} \operatorname{Circ}(f): \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \end{aligned}$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V')$ $(V, A, s, t, \ell) \longmapsto (V', A, (s \div f), (t; f), \ell).$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V');$ $(V, A, s, t, \ell) \longmapsto (V', A, (s \circ f), (t \circ f), \ell).$
fong-spivak-invitation_EQ0214	299	■ 0.401	$\begin{aligned} \Lambda(\text{monoid}, \text{nothing}) &= \Lambda(\text{monoid}, \text{this book}) \\ \Lambda(\text{preorder}, \text{nothing}) &= \Lambda(\text{preorder}, \text{this book}) \\ \Lambda(\text{category}, \text{nothing}) &= \Lambda(\text{category}, \text{this book}) \end{aligned}$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $\Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{preorder}, \text{this book})$ $\Lambda(\text{category}, \text{nothing}) = \Lambda(\text{category}, \text{this book}) = \text{true}$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $\Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{preorder}, \text{this book})$ $\Lambda(\text{category}, \text{nothing}) = \Lambda(\text{category}, \text{this book}) = \text{true}$
fong-spivak-invitation_EQ0024	59	■ 0.473	$\left.\left.\frac{x}{y}\right z\right =\frac{x}{y \otimes z}$	$\frac{\frac{x}{y}}{z} = \frac{x \otimes y}{z} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right. \right.$	$\frac{\frac{x}{y}}{z} = \frac{x \otimes y}{z} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right. \right.$
fong-spivak-invitation_EQ0255	320	■ 0.480	$\operatorname{id}_{\{A\}}+\operatorname{id}_{\{B\}}=\left[\operatorname{id}_{\{A\}}\circ\iota_A, \operatorname{id}_{\{B\}}\circ\iota_B\right]=\left[\iota_A, \iota_B\right]=\operatorname{id}_{A+B}.$	$\operatorname{id}_A + \operatorname{id}_B = [\operatorname{id}_A \circ \iota_A, \operatorname{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \operatorname{id}_{A+B}.$	$\operatorname{id}_A + \operatorname{id}_B = [\operatorname{id}_A \circ \iota_A, \operatorname{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \operatorname{id}_{A+B}.$
fong-spivak-invitation_EQ0210	296	■ 0.483	$H:=\stackrel{4}{\leftarrow} \stackrel{d}{\rightarrow} \stackrel{5}{\rightarrow} \circ e$	$H := \stackrel{4}{\leftarrow} \stackrel{d}{\rightarrow} \stackrel{5}{\rightarrow} \circ e$	$H := \stackrel{4}{\leftarrow} \stackrel{d}{\rightarrow} \stackrel{5}{\rightarrow} \circ e$
fong-spivak-invitation_EQ0186	284	■ 0.486	$\begin{aligned} g(\text{false})+g(\text{false}) &= \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}), \\ g(\text{false})+g(\text{true}) &= \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}), \\ g(\text{true})+g(\text{false}) &= 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}), \\ g(\text{true})+g(\text{true}) &= 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}). \end{aligned}$	$g(\text{ false }) + g(\text{ false }) = \infty + \infty \geq \infty = g(\text{ false } \wedge \text{ false }),$ $g(\text{ false }) + g(\text{ true }) = \infty + 0 \geq \infty = g(\text{ false } \wedge \text{ true }),$ $g(\text{ true }) + g(\text{ false }) = 0 + \infty \geq \infty = g(\text{ true } \wedge \text{ false }),$ $g(\text{ true }) + g(\text{ true }) = 0 + 0 \geq 0 = g(\text{ true } \wedge \text{ true })$	$g(\text{false}) + g(\text{false}) = \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}),$ $g(\text{false}) + g(\text{true}) = \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}),$ $g(\text{true}) + g(\text{false}) = 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}),$ $g(\text{true}) + g(\text{true}) = 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}).$

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fong-spivak-invitation_EQ0092	143	■ 0.504	$(\backslash\Phi\ ;\ \backslash\Psi)\ ;\ \backslash\Upsilon=\backslash\Phi\ ;(\backslash\Psi\ ;\ \backslash\Upsilon)\ .$	$(\Phi;\Psi);\Upsilon = \Phi;(\Psi;\Upsilon).$	$(\Phi \circ \Psi) \circ \Upsilon = \Phi \circ (\Psi \circ \Upsilon).$
fong-spivak-invitation_EQ0077	122	■ 0.507	$\left.\frac{\mathbf{1}}{\text{Em}_6}\right\lvert$	<i>(not rendered)</i>	$\frac{\mathbf{1}}{\text{Em}_6}$
fong-spivak-invitation_EQ0071	118	■ 0.519	$\text{Set} \underset{(-)^B}{\overset{-\times B}{\rightleftarrows}} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B).$	$\text{Set} \underset{(-)^B}{\overset{-\times B}{\rightleftarrows}} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B).$	$\text{Set} \overset{-\times B}{\underset{(-)^B}{\rightleftarrows}} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B).$
fong-spivak-invitation_EQ0206	293	■ 0.529	$F(a):=a^{\prime}, \quad \backslash\text{quad } G(a):=a^{\prime}, \quad \backslash\text{quad } F(b):=b^{\prime}, \quad \backslash\text{quad } G(b):=b^{\prime}, \quad \backslash\text{quad } F(f):=f_1, \quad \backslash\text{quad } G(f):=f_2.$	$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \quad \text{and } G(f) := f_2.$	$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \text{ and } G(f) := f_2.$
fong-spivak-invitation_EQ0074	121	■ 0.559	$\Sigma_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set} \quad \text{and} \quad \Pi_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set}.$	$\Sigma_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set} \quad \text{and} \quad \Pi_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set}.$	$\Sigma_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set} \quad \text{and} \quad \Pi_!:\mathcal{C}\text{-}\text{Inst}\rightarrow\text{Set}.$
fong-spivak-invitation_EQ0239	313	■ 0.561	$c_n:=-c:(1+-c):(1+1+-c):\ldots:(1+\cdots+1+<c):1\rightarrow n.$	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \dots : (1 + \dots + 1 + < c) : 1 \rightarrow n.$	$c_n := -< \circ (1 + -<) \circ (1 + 1 + -<) \circ \dots \circ (1 + \dots + 1 + -<) : 1 \rightarrow n.$
fong-spivak-invitation_EQ0073 (3.14)	121	■ 0.576	$!:\mathcal{C}\rightarrow\mathbf{1}.$	$!:\mathcal{C}\rightarrow\mathbf{1}.$	$!:\mathcal{C}\rightarrow\mathbf{1}.$
fong-spivak-invitation_EQ0025	59	■ 0.583	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$
fong-spivak-invitation_EQ0132	206	■ 0.584	$\left\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\right\}.$	$\{(v,d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$	$\{(v,d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$
fong-spivak-invitation_EQ0057	98	■ 0.592	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f\circ h\}.$
fong-spivak-invitation_EQ0123 (5.14)	189	■ 0.602	$\mathbf{SFG}_R^+ := \text{Free}\left(G_R \sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+ := \text{Free}\left(G_R \sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+ := \text{Free}\left(G_R \sqcup G_R^{\text{op}}\right).$

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fong-spivak-invitation-EQ0221	301	■ 0.622	$\begin{aligned} \Phi: \Psi &= (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \end{aligned}$	$\Phi : \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3)$ $= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3$ $= (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3$ $= \Phi * M_\Psi * M_Z^3$	$\Phi \circ \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3)$ $= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3$ $= (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3$ $= \Phi * M_\Psi * M_Z^3$
fong-spivak-invitation-EQ0010 (1.7)	44	■ 0.623	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-invitation-EQ0155 (7.7)	242	■ 0.623	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-invitation-EQ0156	247	■ 0.628	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$
fong-spivak-invitation-EQ0124	190	■ 0.641	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-invitation-EQ0080 (4.2)	134	■ 0.641	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$
fong-spivak-invitation-EQ0097	150	■ 0.644	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\}$ $= (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\}$ $= (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\}$ $= (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$
fong-spivak-invitation-EQ0224	305	■ 0.649	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$

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fong-spivak-invitation_EQ0055 (3.5)	96	0.649	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$2 := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$2 := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$
fong-spivak-invitation_EQ0107 (5.4)	167	0.670	$G + G^{\prime} := \left(\left(V \sqcup V^{\prime} \right), [\text{in}, \text{in}^{\prime}], [\text{out}, \text{out}^{\prime}], (\iota \sqcup \iota^{\prime}) \right)$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in}'], [\text{out}, \text{out}'], (\iota \sqcup \iota'))$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in}'], [\text{out}, \text{out}'], (\iota \sqcup \iota'))$
fong-spivak-invitation_EQ0151 (7.3)	239	0.683	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^{\mathcal{C}})$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^{\mathcal{C}})$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^{\mathcal{C}})$
fong-spivak-invitation_EQ0191	288	0.683	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0$
fong-spivak-invitation_EQ0200	291	0.687	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ j, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation_EQ0089	142	0.688	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}$
fong-spivak-invitation_EQ0170 (7.18)	263	0.706	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$
fong-spivak-invitation_EQ0061 (3.10)	113	0.708	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \begin{array}{ c } \hline \begin{array}{ccc} \text{Arrow} & \xrightarrow[\text{target}]{\text{source}} & \text{Vertex} \\ \bullet & & \bullet \end{array} \\ \hline \text{no equations} \end{array}$
fong-spivak-invitation_EQ0018	49	0.726	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}$
fong-spivak-invitation_EQ0149 (7.2)	234	0.727	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0)$
fong-spivak-invitation_EQ0082	135	0.730	$\Phi: x^{\operatorname{op}} \times y \rightarrow \mathcal{V}$	$\Phi: x^{\operatorname{op}} \times y \rightarrow \mathcal{V}$	$\Phi: \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$
fong-spivak-invitation_EQ0248	316	0.734	$\mathcal{B}(g \circ (h^{\operatorname{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}$	$\mathcal{B}(g \circ (h^{\operatorname{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}$	$\mathcal{B}(g \circ (h^{\operatorname{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}$
fong-spivak-invitation_EQ0114	178	0.740	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$
fong-spivak-invitation_EQ0138	220	0.748	$C := \{ \text{light, switch, battery} \} \sqcup \{ x \Omega \mid x \in \mathbb{R}^+ \}$	$C := \{ \text{light, switch, battery} \} \sqcup \{ x \Omega \mid x \in \mathbb{R}^+ \}$	$C := \{ \text{light, switch, battery} \} \sqcup \{ x \Omega \mid x \in \mathbb{R}^+ \}$
fong-spivak-invitation_EQ0143 (6.21)	226	0.754	$\begin{aligned} \circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \\ \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t) \end{aligned}$	$\circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t)$	$\circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t)$
fong-spivak-invitation_EQ0198	290	0.762	$\stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{v_n}{\bullet}$	$\stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{v_n}{\bullet}$	$\stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{v_n}{\bullet}$
fong-spivak-invitation_EQ0209	296	0.772	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$
fong-spivak-invitation_EQ0160	253	0.775	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}$

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fong-spivak-invitation_E00047	83	■ 0.847	$2 \mathrm{H}_{\mathrm{2}} \mathrm{~} \rightarrow \mathrm{~} \left(2 \mathrm{~} \mathrm{Na} \mathrm{~} \multimap \mathrm{~} \left(2 \mathrm{~} \mathrm{NaOH} + \mathrm{H}_{\mathrm{2}} \right) \right) \mathrm{~} .$	$2\mathrm{H}_2\mathrm{O} \rightarrow (2\mathrm{Na} \multimap (2\mathrm{NaOH} + \mathrm{H}_2)).$	$2\mathrm{H}_2\mathrm{O} \rightarrow (2\mathrm{Na} \multimap (2\mathrm{NaOH} + \mathrm{H}_2)).$
fong-spivak-invitation_E00088 (4.8)	142	■ 0.851	$\mathrm{U}_{\mathrm{U}_{\mathrm{mathcal{X}}}}(x,y) := \mathfrak{U}(x,y) \mathrm{~} .$	$\mathrm{U}_{\mathcal{X}}(x,y) := \mathfrak{X}(x,y).$	$\mathrm{U}_{\mathcal{X}}(x,y) := \mathfrak{X}(x,y).$
fong-spivak-invitation_E00090 (4.9)	142	■ 0.868	$\begin{aligned} \Phi(p,q) \, \&= I \otimes \Phi(p,q) \\ \leq \mathcal{P}(p,p) \otimes \Phi(p,q) \\ \bigvee_{p_1 \in P} \left(\mathcal{P}(p,p_1) \otimes \Phi(p_1,q) \right) &= (\mathrm{U}_{\mathcal{P}}; \Phi)(p,q) . \end{aligned}$	$\begin{aligned} \Phi(p,q) &= I \otimes \Phi(p,q) \leq \mathcal{P}(p,p) \otimes \Phi(p,q) \\ &\leq \bigvee_{p_1 \in P} (\mathcal{P}(p,p_1) \otimes \Phi(p_1,q)) = (\mathrm{U}_{\mathcal{P}}; \Phi)(p,q) . \end{aligned}$	$\begin{aligned} \Phi(p,q) &= I \otimes \Phi(p,q) \leq \mathcal{P}(p,p) \otimes \Phi(p,q) \\ &\leq \bigvee_{p_1 \in P} (\mathcal{P}(p,p_1) \otimes \Phi(p_1,q)) = (\mathrm{U}_{\mathcal{P}} \circ \Phi)(p,q) . \end{aligned}$
fong-spivak-invitation_E00099	152	■ 0.880	$(f \times g) \, ; \, \left(f^{\mathrm{~} \prime} \times g^{\mathrm{~} \prime} \right) \, = \, (f \, ; \, g) \, \times \, \left(f^{\mathrm{~} \prime} \, ; \, g^{\mathrm{~} \prime} \right) \mathrm{~} .$	$(f \times g); (f' \times g') = (f; g) \times (f'; g') .$	$(f \times g) \circ (f' \times g') = (f \circ g) \times (f' \circ g') .$
fong-spivak-invitation_E00167	260	■ 0.881	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s) .$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s) .$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s) .$
fong-spivak-invitation_E00129 (6.2)	200	■ 0.882	$[f,g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A, \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$	$[f,g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A, \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$	$[f,g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A, \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$
fong-spivak-invitation_E00144	228	■ 0.883	$\begin{aligned} (g \circ f) \left(x_1, \dots, x_{i-1}, \dots, x_n \right) &= g \left(f \left(x_1, \dots, x_{i-1}, \dots, x_n \right) \right) \\ &= g \left(x_1, \dots, x_{i-1}, f \left(x_i, \dots, x_n \right) \right) \end{aligned}$	$\begin{aligned} (g \circ f) \left(x_1, \dots, x_{i-1}, x_1, \dots, x_n \right) &= g \left(x_1, \dots, x_{i-1}, f \left(x_1, \dots, x_n \right) \right) \\ &= g \left(x_1, \dots, x_{i-1}, f \left(x_1, \dots, x_n \right) \right) \end{aligned}$	$\begin{aligned} (g \circ f) \left(x_1, \dots, x_{i-1}, w_1, \dots, w_m, x_{i+1}, \dots, x_n \right) &= g \left(x_1, \dots, x_{i-1}, f \left(w_1, \dots, w_m \right), x_{i+1}, \dots, x_n \right) \\ &= g \left(x_1, \dots, x_{i-1}, f \left(w_1, \dots, w_m \right), x_{i+1}, \dots, x_n \right) \end{aligned}$
fong-spivak-invitation_E00119	188	■ 0.887	$\mathrm{B}(-\mathrm{C}) = \{ (x, (x, x)) \mid x \in R \} .$	$\mathrm{B}(-\mathrm{C}) = \{ (x, (x, x)) \mid x \in R \} .$	$\mathrm{B}(\neg) = \{ (x, (x, x)) \mid x \in R \} .$
fong-spivak-invitation_E00165 (7.16)	257	■ 0.892	$(U \Rightarrow V) := \bigcup_{\{R \in \mathbf{Op} \mid R \cap U \subseteq V\}} R .$	$(U \Rightarrow V) := \bigcup_{\{R \in \mathbf{Op} \mid R \cap U \subseteq V\}} R .$	$(U \Rightarrow V) := \bigcup_{\{R \in \mathbf{Op} \mid R \cap U \subseteq V\}} R .$
fong-spivak-invitation_E00153	241	■ 0.910	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X, \\ \text{false} & \text{otherwise} . \end{cases}$	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X, \\ \text{false} & \text{otherwise} . \end{cases}$	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X, \\ \text{false} & \text{otherwise} . \end{cases}$
fong-spivak-invitation_E00122 (5.13)	189	■ 0.912	$B_1; B_2 := \{ (x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2 \} .$	$B_1; B_2 := \{ (x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2 \} .$	$B_1 \circ B_2 := \{ (x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2 \} .$
fong-spivak-invitation_E00235	312	■ 0.913	$A+B = \begin{pmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{pmatrix} .$	$A+B = \begin{pmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{pmatrix} .$	$A+B = \begin{pmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{pmatrix} .$
fong-spivak-invitation_E00005	36	■ 0.924	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)) \quad \text{and} \quad q = f(g(q)) .$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)) .$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)) .$

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fong-spivak-invitation_E00193	289	■ 0.936	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z) \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z) \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$
fong-spivak-invitation_E00093	144	■ 0.937	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$
fong-spivak-invitation_E00174 (7.19)	268	■ 0.937	$\begin{aligned} &\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ &\exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)). \end{aligned}$	$\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$	$\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$
fong-spivak-invitation_E00079	127	■ 0.943	$\begin{aligned} \lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \} \end{aligned}$	$\lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \}$	$\lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \}$
fong-spivak-invitation_E00226	305	■ 0.948	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\text{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\text{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{U_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\text{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times U_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$
fong-spivak-invitation_E00145 (6.22)	228	■ 0.950	$(g \circ i f) : X_1 \times \dots \times X_{i-1} \times W_1 \times \dots \times W_m \times X_{i+1} \times \dots \times X_n \longrightarrow Y,$	$(g \circ i f) : X_1 \times \dots \times X_{i-1} \times W_1 \times \dots \times W_m \times X_{i+1} \times \dots \times X_n \longrightarrow Y,$	$(g \circ_i f) : X_1 \times \dots \times X_{i-1} \times W_1 \times \dots \times W_m \times X_{i+1} \times \dots \times X_n \longrightarrow Y,$
fong-spivak-invitation_E00207	293	■ 0.954	$(F ; G)(f ; g) = G(F(f) ; F(g)) = G(F(f)) ; G(F(g))$	$(F ; G)(f ; g) = G(F(f) ; F(g)) = G(F(f)) ; G(F(g))$	$(F \circledast G)(f \circledast g) = G(F(f) \circledast F(g)) = G(F(f)) \circledast G(F(g))$
fong-spivak-invitation_E00048 (2.24)	84	■ 0.954	$\left(v \otimes \bigvee_{a \in A} a \right) \cong \bigvee_{a \in A} (v \otimes a).$	$\left(v \otimes \bigvee_{a \in A} a \right) \cong \bigvee_{a \in A} (v \otimes a).$	$\left(v \otimes \bigvee_{a \in A} a \right) \cong \bigvee_{a \in A} (v \otimes a).$

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fong-spivak-invitation_EQ0219	300	■ 0.957	$\begin{aligned} \Phi &= M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \end{aligned}$	$\Phi = M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ = \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$	$\Phi = M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ = \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$
fong-spivak-invitation_EQ0072	120	■ 0.959	$\Sigma_{\{I\}}(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$	$\Sigma_F(I)(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$	$\Sigma_F(I)(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$
fong-spivak-invitation_EQ0103	159	■ 0.962	$\eta_X: \mathbf{1} \times \mathcal{X}^{\text{op}} \times \mathcal{X} \rightarrow \mathcal{V};$	$\eta_X: \mathbf{1} \times \mathcal{X}^{\text{op}} \times \mathcal{X} \rightarrow \mathcal{V};$	$\eta_X: \mathbf{1} \times \mathcal{X}^{\text{op}} \times \mathcal{X} \rightarrow \mathcal{V};$
fong-spivak-invitation_EQ0046	83	■ 0.964	$a + c \rightarrow d \quad \text{iff} \quad a \rightarrow (c \multimap d).$	$a + c \rightarrow d \quad \text{iff} \quad a \rightarrow (c \multimap d).$	$a + c \rightarrow d \quad \text{iff} \quad a \rightarrow (c \multimap d).$
fong-spivak-invitation_EQ0110	176	■ 0.965	$\begin{aligned} \dot{x} + 3\ddot{y} - 2z &= 0, \\ \ddot{y} + 5\dot{z} &= 0. \end{aligned}$	$\begin{aligned} \dot{x} + 3\ddot{y} - 2z &= 0, \\ \ddot{y} + 5\dot{z} &= 0. \end{aligned}$	$\begin{aligned} \dot{x} + 3\ddot{y} - 2z &= 0, \\ \ddot{y} + 5\dot{z} &= 0. \end{aligned}$
fong-spivak-invitation_EQ0172	265	■ 0.965	$\mathbb{R} \cong \{[d, u] \in \mathbb{R} \mid d = u\}.$	$\mathbb{R} \cong \{[d, u] \in \mathbb{R} \mid d = u\}.$	$\mathbb{R} \cong \{[d, u] \in \mathbb{R} \mid d = u\}.$
fong-spivak-invitation_EQ0216	300	■ 0.965	$X(x', x) \otimes y(y, y') \leq \Phi(x, y) \multimap \Phi(x', y').$	$X(x', x) \otimes y(y, y') \leq \Phi(x, y) \multimap \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \mathcal{Y}(y, y') \leq \Phi(x, y) \multimap \Phi(x', y').$
fong-spivak-invitation_EQ0250	316	■ 0.966	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}.$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}.$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}.$
fong-spivak-invitation_EQ0168	261	■ 0.967	$\forall (t : T). p(s, t).$	$\forall (t : T). p(s, t).$	$\forall (t : T). p(s, t).$
fong-spivak-invitation_EQ0007	40	■ 0.967	$f(a) \vee f(b) \supseteq f(a \vee b).$	$f(a) \vee f(b) \not\supseteq f(a \vee b).$	$f(a) \vee f(b) \not\supseteq f(a \vee b).$
fong-spivak-invitation_EQ0101	158	■ 0.969	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y').$	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y').$	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y').$
fong-spivak-invitation_EQ0094 (4.11)	145	■ 0.970	$\mathcal{P}(p, G(q)) \cong Q(F(p), q).$	$\mathcal{P}(p, G(q)) \cong Q(F(p), q).$	$\mathcal{P}(p, G(q)) \cong Q(F(p), q).$
fong-spivak-invitation_EQ0213	299	■ 0.971	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in X \text{ and } d_2 \in Y\},$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in X \text{ and } d_2 \in Y\},$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in X \text{ and } d_2 \in Y\},$
fong-spivak-invitation_EQ0259	328	■ 0.972	$p; h_3 = r; g; h_3 = r; h_2; g' = q; f'; g',$	$p; h_3 = r; g; h_3 = r; h_2; g' = q; f'; g',$	$p \circ h_3 = r \circ g \circ h_3 = r \circ h_2 \circ g' = q \circ f' \circ g',$
fong-spivak-invitation_EQ0152 (7.5)	241	■ 0.972	$\Omega_{\text{Set}} := \mathbb{B} = \{\text{true}, \text{false}\}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{\text{true}, \text{false}\}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{\text{true}, \text{false}\}.$
fong-spivak-invitation_EQ0068 (3.13)	117	■ 0.979	$f(p) \leq q \text{ iff } p \leq g(q).$	$f(p) \leq q \text{ iff } p \leq g(q).$	$f(p) \leq q \text{ iff } p \leq g(q).$

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fong-spivak-invitation_EQ0111	176	■ 0.979	$\begin{array}{r} D \ x+3 \ D^{\{2\}} \ y-2 \ z=0 \ \backslash \ D^{\{2\}} \ y+5 \ D \ z=0 \\ \end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0 \\ D^2y+5Dz=0 \end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0, \\ D^2y+5Dz=0, \end{array}$
fong-spivak-invitation_EQ0104 (5.2)	163	■ 0.979	$i \longmapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m'+g(i) & \text{if } m+1 \leq i \leq m+n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m'+g(i) & \text{if } m+1 \leq i \leq m+n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m'+g(i) & \text{if } m+1 \leq i \leq m+n. \end{cases}$
fong-spivak-invitation_EQ0016	47	■ 0.980	$f_{\{!\}}\!\left(A^{\{\prime\}}\!\right)\!:=\!\left\{b\in B\mid \text{there exists } a\in A^{\{\prime\}}\!\text{ such that } f(a)=b\right\}.$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}.$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}.$
fong-spivak-invitation_EQ0076	122	■ 0.982	$\begin{array}{c c} 1 & \\ \hline \text{Bob-Grace-Pat-Emmy} & \\ \text{Sue-Doug} & \end{array}$	$\begin{array}{c c} 1 & \\ \hline \text{Bob-Grace-Pat-Emmy} & \\ \text{Sue-Doug} & \end{array}$	$\begin{array}{c c} \mathbf{1} & \\ \hline \text{Bob-Grace-Pat-Emmy} & \\ \text{Sue-Doug} & \end{array}$
fong-spivak-invitation_EQ0034	67	■ 0.983	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$
fong-spivak-invitation_EQ0131	206	■ 0.985	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$
fong-spivak-invitation_EQ0247	316	■ 0.986	$\left\{ (x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y) \right\}.$	$\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$	$\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$
fong-spivak-invitation_EQ0017	47	■ 0.988	$f_{\{*\}}\!\left(A^{\{\prime\}}\!\right)\!:=\!\left\{b\in B\mid \text{for all } a\text{ such that } f(a)=b, \text{ we have } a\in A^{\{\prime\}}\!\right\}.$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}.$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}.$
fong-spivak-invitation_EQ0069	118	■ 0.988	$\alpha_{c,d} : \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d} : \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d} : \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$
fong-spivak-invitation_EQ0228	307	■ 0.992	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m, \\ i-m & \text{if } m+1 \leq i. \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m, \\ i-m & \text{if } m+1 \leq i. \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m, \\ i-m & \text{if } m+1 \leq i. \end{cases}$
fong-spivak-invitation_EQ0126 (and)	191	■ 0.992	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$
fong-spivak-invitation_EQ0161 (7.13)	255	■ 0.992	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$
fong-spivak-invitation_EQ0042	79	■ 0.993	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2)),$
fong-spivak-invitation_EQ0164 (7.15)	257	■ 0.994	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$
fong-spivak-invitation_EQ0171	265	■ 0.994	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$

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fong-spivak-invitation_EQ0192	289	■ 0.994	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y) \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$
fong-spivak-invitation_EQ0113 (5.7)	178	■ 0.995	$M ; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c).$	$M; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c).$	$M \circ N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c).$
fong-spivak-invitation_EQ0056 (3.6)	96	■ 0.995	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$
fong-spivak-invitation_EQ0195	290	■ 0.996	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$
fong-spivak-invitation_EQ0002 (1.5)	23	■ 0.996	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$
fong-spivak-invitation_EQ0189	288	■ 0.996	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$
fong-spivak-invitation_EQ0136	217	■ 0.996	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$
fong-spivak-invitation_EQ0009	42	■ 0.997	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-invitation_EQ0053	87	■ 0.997	$\left(\begin{array}{cc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array} \right) * \left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array} \right) = \left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array} \right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array} \right) * \left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array} \right) = \left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array} \right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array} \right) * \left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array} \right) = \left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array} \right)$
fong-spivak-invitation_EQ0231	310	■ 0.997	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$
fong-spivak-invitation_EQ0049	86	■ 0.997	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v).$

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fong-spivak-invitation_EQ0215	300	■ 0.997	$\left(\mathrm{X}^{\mathrm{op}} \times \mathrm{Y}\right) \left((x, y), \left(x', y'\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x', y'\right)\right).$	$(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y})((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y')).$	$(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y})((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y')).$
fong-spivak-invitation_EQ0108	169	■ 0.998	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$	$a \curvearrowright \bullet \curvearrowleft b$
fong-spivak-invitation_EQ0166 (7.17)	260	■ 0.998	$\left(\left \Omega^S\right ,\leq^S\right).$	$(\Omega^S , \leq^S).$	$(\Omega^S , \leq^S).$
fong-spivak-invitation_EQ0011	45	■ 0.998	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-invitation_EQ0106	166	■ 0.998	$\iota^{\mathrm{\prime}}(x)=\begin{cases}\iota(x) & \text{if } \iota(x) \in I, \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'.\end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{if } \iota(x) \in I, \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'. \end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{if } \iota(x) \in I, \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'. \end{cases}$
fong-spivak-invitation_EQ0008 (1.6)	41	■ 0.999	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$
fong-spivak-invitation_EQ0212	298	■ 0.999	$\left(c_1, d_1\right) \xrightarrow{\left(f_1, g_1\right)}\left(c_2, d_2\right) \xrightarrow{\left(f_2, g_2\right)}\left(c_3, d_3\right) \xrightarrow{\left(f_3, g_3\right)}\left(c_4, d_4\right).$	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$
fong-spivak-invitation_EQ0232	310	■ 0.999	$G:=\left\{\rho_{m,n}:m\rightarrow n\mid m,n\in\mathbb{N}\right\},\quad s\left(\rho_{m,n}\right):=m,\quad t\left(\rho_{m,n}\right):=n;$	$G:=\{\rho_{m,n}:m\rightarrow n\mid m,n\in\mathbb{N}\},\quad s(\rho_{m,n}):=m,\quad t(\rho_{m,n}):=n;$	$G:=\{\rho_{m,n}:m\rightarrow n\mid m,n\in\mathbb{N}\},\quad s(\rho_{m,n}):=m,\quad t(\rho_{m,n}):=n;$
fong-spivak-invitation_EQ0253	318	■ 0.999	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \cdots + 1_R}_{m \text{ summands}} + \underbrace{\cdots + 1_R + \cdots + 1_R}_{n \text{ summands}}) *_R (\underbrace{1_R + \cdots + 1_R}_{m \text{ summands}} + \underbrace{\cdots + 1_R + \cdots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) + \cdots + 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}})}_{m \text{ summands}} \\ &= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \cdots + 1_R}_{m \text{ summands}} *_R \underbrace{(1_R + \cdots + 1_R)_{n \text{ summands}}}_{n \text{ summands}}) \\ &= 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) + \cdots + 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \cdots + 1_R}_{m \text{ summands}} *_R \underbrace{(1_R + \cdots + 1_R)_{n \text{ summands}}}_{n \text{ summands}}) \\ &= 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) + \cdots + 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$
fong-spivak-invitation_EQ0146	230	■ 0.999	$(f \circ i g) := (\mathrm{id} \otimes, \ldots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \ldots, \otimes \mathrm{id}); f,$	$(f \circ i g) := (\mathrm{id} \otimes, \ldots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \ldots, \otimes \mathrm{id}); f,$	$(f \circ_i g) := (\mathrm{id} \otimes, \ldots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \ldots, \otimes \mathrm{id}) \circ f,$
fong-spivak-invitation_EQ0015	47	■ 0.999	$f\left(g\left(q_{\{0\}}\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_{\{0\}}\right) \leq g(q)\right\} \leq \bigwedge\left\{q_{\{0\}}=q_{\{0\}}\right\},$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$
fong-spivak-invitation_EQ0081 (4.3)	134	■ 0.999	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$

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fong-spivak-invitation_EQ0091 (4.10)	142	■ 0.999	$\begin{aligned} \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) &= \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes I \\ &\leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) . \end{aligned}$	$\mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) = \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes I \\ \leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) = \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \\ \leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$
fong-spivak-invitation_EQ0157	248	■ 0.999	$\mathbf{0\ p_{\{1\}}}:=\{\text{varnothing},\{1\},\ X\}\quad\text{and}\quad\mathbf{0\ p_{\{2\}}}:=\{\text{varnothing},\{2\},\ X\}\ .$	$\mathbf{Op_1} := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op_2} := \{\emptyset, \{2\}, X\}.$	$\mathbf{Op_1} := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op_2} := \{\emptyset, \{2\}, X\} .$
fong-spivak-invitation_EQ0183	276	■ 0.999	$f^{\{*\}}(P)=\{\{x\in\mathbb{Z}\mid x\leq 0\},\{x\in\mathbb{Z}\mid x\geq 1\}\}\quad\text{and}\quad f^{\{*\}}(Q)=\{\{0\},\{x\in\mathbb{Z}\mid x\neq 0\}\}\ .$	$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$	$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$
fong-spivak-invitation_EQ0245	316	■ 0.999	$B+C:=\left\{(w,\ y,\ x,\ z)\in R^{m+p}\times R^{n+q}\mid (w,\ x)\in B\ \text{and}\ (y,\ z)\in C\right\}\ .$	$B+C:=\left\{(w,y,x,z)\in R^{m+p}\times R^{n+q}\mid (w,x)\in B\ \text{and}\ (y,z)\in C\right\}\ .$	$B+C:=\{(w,y,x,z)\in R^{m+p}\times R^{n+q}\mid (w,x)\in B\ \text{and}\ (y,z)\in C\}.$
fong-spivak-invitation_EQ0039	78	■ 0.999	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c,c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c,c) \quad (\text{definition of } \mathcal{C}_f); \end{array}$	$\begin{array}{ll} I_W \leq f(I_V) & (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c,c)) & (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c,c) & (\text{definition of } \mathcal{C}_f); \end{array}$	$\begin{array}{ll} I_W \leq f(I_V) & (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) & (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) & (\text{definition of } \mathcal{C}_f); \end{array}$
fong-spivak-invitation_EQ0142 (6.16)	221	■ 0.999	$\begin{aligned} \psi_{V,V'}\colon \operatorname{Circ}(V)\times \operatorname{Circ}(V')\longrightarrow \operatorname{Circ}(V+V'); \\ ((V,A,s,t,\ell),(V',A',s',t',\ell'))\longmapsto (V+V',A+A',s+s',t+t',[\ell,\ell']). \end{aligned}$	$\begin{array}{l} \psi_{V,V'}\colon \operatorname{Circ}(V)\times \operatorname{Circ}(V')\longrightarrow \operatorname{Circ}(V+V'); \\ ((V,A,s,t,\ell),(V',A',s',t',\ell'))\longmapsto (V+V',A+A',s+s',t+t',[\ell,\ell']). \end{array}$	$\begin{array}{l} \psi_{V,V'}\colon \operatorname{Circ}(V)\times \operatorname{Circ}(V')\longrightarrow \operatorname{Circ}(V+V'); \\ ((V,A,s,t,\ell),(V',A',s',t',\ell'))\longmapsto (V+V',A+A',s+s',t+t',[\ell,\ell']). \end{array}$
fong-spivak-invitation_EQ0052 (2.26)	87	■ 1.000	$(M\ast N)(x,z) := \bigvee_{y\in Y} M(x,y)\otimes N(y,z)$	$(M\ast N)(x,z) := \bigvee_{y\in Y} M(x,y)\otimes N(y,z).$	$(M\ast N)(x,z) := \bigvee_{y\in Y} M(x,y)\otimes N(y,z).$
fong-spivak-invitation_EQ0054	87	■ 1.000	$I_X(x,y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$	$I_X(x,y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$	$I_X(x,y) := \begin{cases} I & \text{if } x = y, \\ 0 & \text{if } x \neq y. \end{cases}$
fong-spivak-invitation_EQ0063	113	■ 1.000	$\alpha_{\text{Vertex}}\colon G(\text{Vertex})\rightarrow H(\text{Vertex})\quad\text{and}\quad\alpha_{\text{Arrow}}\colon G(\text{Arrow})\rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}}\colon G(\text{Vertex})\rightarrow H(\text{Vertex})\quad\text{and}\quad\alpha_{\text{Arrow}}\colon G(\text{Arrow})\rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}}\colon G(\text{Vertex})\rightarrow H(\text{Vertex})\quad\text{and}\quad\alpha_{\text{Arrow}}\colon G(\text{Arrow})\rightarrow H(\text{Arrow}),$
fong-spivak-invitation_EQ0067 (3.12)	116	■ 1.000	$\text{Gr}\xrightarrow{F}\text{DDS}\xrightarrow{I}\text{Set}.$	$\text{Gr}\xrightarrow{F}\text{DDS}\xrightarrow{I}\text{Set}.$	$\mathbf{Gr}\xrightarrow{F}\mathbf{DDS}\xrightarrow{I}\mathbf{Set}.$
fong-spivak-invitation_EQ0181	273	■ 1.000	$1\leq 1,\quad 1\leq 2,\quad 1\leq 3,\quad 2\leq 2,\quad 2\leq 3,\quad 3\leq 3,\quad 4\leq 4.$	$1\leq 1,\quad 1\leq 2,\quad 1\leq 3,\quad 2\leq 2,\quad 2\leq 3,\quad 3\leq 3,\quad 4\leq 4.$	$1\leq 1,\quad 1\leq 2,\quad 1\leq 3,\quad 2\leq 2,\quad 2\leq 3,\quad 3\leq 3,\quad 4\leq 4.$
fong-spivak-invitation_EQ0220	301	■ 1.000	$=\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right).$	$=\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right).$	$=\begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}.$

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fong-spivak-invitation_EQ0258	324	■ 1.000	$\begin{aligned} \operatorname{varphi}_{S^{\prime}, T^{\prime}}^{\prime} \left(\operatorname{im}_f \times \operatorname{im}_g \right) (A \times B) &= \operatorname{varphi}_{S^{\prime}, T^{\prime}}^{\prime} (\{ f(a) \mid a \in A \} \times \{ g(b) \mid b \in B \}) \\ &= \{ (f(a), g(b)) \mid a \in A, b \in B \} \\ &= \operatorname{im}_{f \times g} (A \times B) \\ &= \operatorname{im}_{f \times g} (\varphi_{S, T} (A, B)). \end{aligned}$	$\begin{aligned} \varphi_{S', T'} ((\operatorname{im}_f \times \operatorname{im}_g) (A \times B)) &= \varphi_{S', T'} (\{ f(a) \mid a \in A \} \times \{ g(b) \mid b \in B \}) \\ &= \{ (f(a), g(b)) \mid a \in A, b \in B \} \\ &= \operatorname{im}_{f \times g} (A \times B) \\ &= \operatorname{im}_{f \times g} (\varphi_{S, T} (A, B)). \end{aligned}$	$\begin{aligned} \varphi_{S', T'} ((\operatorname{im}_f \times \operatorname{im}_g) (A \times B)) &= \varphi_{S', T'} (\{ f(a) \mid a \in A \} \times \{ g(b) \mid b \in B \}) \\ &= \{ (f(a), g(b)) \mid a \in A, b \in B \} \\ &= \operatorname{im}_{f \times g} (A \times B) \\ &= \operatorname{im}_{f \times g} (\varphi_{S, T} (A, B)). \end{aligned}$
fong-spivak-invitation_EQ0001 (1.4)	20	■ 1.000	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B) \text{ .}$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-invitation_EQ0004	26	■ 1.000	$\begin{aligned} f(11) &= a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \\ f(22) &= d, \quad f(23) = d \text{ .} \end{aligned}$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d.$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d.$
fong-spivak-invitation_EQ0006	37	■ 1.000	$\left\{ \left. \frac{1}{n+1} \right n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \left. \frac{1}{n+1} \right n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$
fong-spivak-invitation_EQ0012	45	■ 1.000	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$
fong-spivak-invitation_EQ0013 (1.8)	46	■ 1.000	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \} \text{ ;}$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$
fong-spivak-invitation_EQ0014	47	■ 1.000	$\begin{aligned} p_{\{0\}} \leq \bigwedge \{ g(q) \in P \mid p_{\{0\}} \leq g(q) \} \right) &\cong g \left(\bigwedge \{ q \in Q \mid p_0 \leq g(q) \} \right) \\ &= g \left(f \left(p_{\{0\}} \right) \right) \end{aligned}$	$p_0 \leq \bigwedge \{ g(q) \in P \mid p_0 \leq g(q) \} \cong g \left(\bigwedge \{ q \in Q \mid p_0 \leq g(q) \} \right) = g \left(f \left(p_0 \right) \right),$	$p_0 \leq \bigwedge \{ g(q) \in P \mid p_0 \leq g(q) \} \cong g \left(\bigwedge \{ q \in Q \mid p_0 \leq g(q) \} \right) = g \left(f \left(p_0 \right) \right),$
fong-spivak-invitation_EQ0019 (2.2)	55	■ 1.000	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$
fong-spivak-invitation_EQ0026 (2.8)	60	■ 1.000	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$
fong-spivak-invitation_EQ0027 (2.9)	60	■ 1.000	$t + u \leq y + z \text{ .}$	$t + u \leq y + z.$	$t + u \leq y + z.$
fong-spivak-invitation_EQ0028 (2.10)	60	■ 1.000	$t + u \leq v + w + u \leq v + x + z \leq y + z \text{ .}$	$t + u \leq v + w + u \leq v + x + z \leq y + z.$	$t + u \leq v + w + u \leq v + x + z \leq y + z.$
fong-spivak-invitation_EQ0029	61	■ 1.000	$\mathrm{H}_2 \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4 + 3 \mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$
fong-spivak-invitation_EQ0030	61	■ 1.000	$2 \mathrm{H}_2 \mathrm{O} + 2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH} + \mathrm{H}_2.$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$
fong-spivak-invitation_EQ0031 (2.11)	62	■ 1.000	$\begin{aligned} y + k &\rightarrow y' + k', \quad x + y' \rightarrow z', \quad z' + k' \rightarrow z + k. \\ z^{\prime} &\rightarrow z + k \text{ .} \end{aligned}$	$y + k \rightarrow y' + k', \quad x + y' \rightarrow z', \quad z' + k' \rightarrow z + k.$	$y + k \rightarrow y' + k', \quad x + y' \rightarrow z', \quad z' + k' \rightarrow z + k.$
fong-spivak-invitation_EQ0032 (2.12)	62	■ 1.000	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$
fong-spivak-invitation_EQ0033 (2.14)	63	■ 1.000	$\mathrm{H}_2 \mathrm{O} + \mathrm{NaCl} \rightarrow \text{?} \mathrm{H}_2 \mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow \text{?} \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow \text{?} \mathrm{H}_2\mathrm{O}$
fong-spivak-invitation_EQ0035	69	■ 1.000	$\begin{aligned} d(x) &:= \begin{cases} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{cases} \\ &\quad \text{if } x = \infty, \quad \text{if } x < \infty. \end{aligned}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{cases}$

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fong-spivak-invitation_EQ0036	73	■ 1.000	$d_{\{L\}}(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-invitation_EQ0038 (2.20)	76	■ 1.000	$M_{\{Y\}} := \begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
fong-spivak-invitation_EQ0041 (2.21)	78	■ 1.000	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$
fong-spivak-invitation_EQ0044 (2.23)	82	■ 1.000	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$
fong-spivak-invitation_EQ0045	83	■ 1.000	$a + x \geq y \quad \text{iff} \quad a \geq y - x \quad \text{iff} \quad \max(0, a) \geq \max(0, y - x) \quad \text{iff} \quad a \geq (x \multimap y)$	$a + x \geq y \quad \text{iff} \quad a \geq y - x \quad \text{iff} \quad \max(0, a) \geq \max(0, y - x) \quad \text{iff} \quad a \geq (x \multimap y)$	$a + x \geq y \quad \text{iff} \quad a \geq y - x \quad \text{iff} \quad \max(0, a) \geq \max(0, y - x) \quad \text{iff} \quad a \geq (x \multimap y)$
fong-spivak-invitation_EQ0050	86	■ 1.000	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$
fong-spivak-invitation_EQ0051 (2.25)	87	■ 1.000	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$
fong-spivak-invitation_EQ0058	107	■ 1.000	$f^{\prime}; h^{\prime} = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g^{\prime}; i^{\prime}.$	$f^{\prime}; h^{\prime} = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g^{\prime}; i^{\prime}.$	$f^{\prime}; h^{\prime} = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g^{\prime}; i^{\prime}.$
fong-spivak-invitation_EQ0060 (3.9)	110	■ 1.000	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f); \alpha_d = \alpha_c; G(f).$
fong-spivak-invitation_EQ0078	124	■ 1.000	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$
fong-spivak-invitation_EQ0084 (4.5)	137	■ 1.000	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$
fong-spivak-invitation_EQ0086 (4.7)	140	■ 1.000	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$
fong-spivak-invitation_EQ0087	141	■ 1.000	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$
fong-spivak-invitation_EQ0098	150	■ 1.000	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$
fong-spivak-invitation_EQ0102	158	■ 1.000	$(\Phi \times \Psi)(\left(\left(x_1, y_1\right), \left(x_2, y_2\right)\right) := \Phi\left(x_1, x_2\right) \otimes \Psi\left(y_1, y_2\right).$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$
fong-spivak-invitation_EQ0105	164	■ 1.000	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}.$	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}.$	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}.$
fong-spivak-invitation_EQ0109 (5.5)	170	■ 1.000	$G := \left\{ \rho_{m,n} \mid m, n \in \mathbb{N} \right\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{ \rho_{m,n} \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{ \rho_{m,n} \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$
fong-spivak-invitation_EQ0115	178	■ 1.000	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right).$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right).$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right).$

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fong-spivak-invitation_EQ0117	181	■ 1.000	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) \text{ .}$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$
fong-spivak-invitation_EQ0120 (5.11)	188	■ 1.000	$\mathrm{B}(g) = \left\{ (x, S(g)(x)) \mid x \in R^m \right\} \subseteq R^m \times R^n \text{ .}$	$B(g) = \{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$	$B(g) = \{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$
fong-spivak-invitation_EQ0121 (5.12)	188	■ 1.000	$\mathrm{B}\left(g^{\mathrm{op}}\right) := \left\{ (S(g)(x), x) \mid x \in R^m \right\} \subseteq R^n \times R^m \text{ .}$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m.$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m.$
fong-spivak-invitation_EQ0125	190	■ 1.000	$\mathrm{B}\left(\left(\left(g^{\mathrm{op}}\right) ; h\right) ; h\right) = \left\{ (S(g)(x), S(h)(x)) \mid x \in R^m \right\} \text{ .}$	$B((g^{\mathrm{op}}); h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$	$B((g^{\mathrm{op}}) \circ h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$
fong-spivak-invitation_EQ0127	192	■ 1.000	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt \text{ .}$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$
fong-spivak-invitation_EQ0130	204	■ 1.000	$y(\theta) = y(g(\theta)) = x(f(\theta)) = x(\theta) \text{ .}$	$y(0) = y(g(0)) = x(f(0)) = x(0).$	$y(0) = y(g(0)) = x(f(0)) = x(0).$
fong-spivak-invitation_EQ0133	207	■ 1.000	$\{(v, d) \mid d \in D(v)\} \text{ .}$	$\{(v, d) \mid d \in D(v)\}.$	$\{(v, d) \mid d \in D(v)\}.$
fong-spivak-invitation_EQ0137 (6.15)	219	■ 1.000	$F\left(\left(\left[\iota_N, \iota_P\right]\right)\right)\left(\left(\left[\varphi_N, P\right](s, t)\right)\right) \text{ .}$	$F([\iota_N, \iota_P])(\varphi_{N,P}(s, t)).$	$F([\iota_N, \iota_P])(\varphi_{N,P}(s, t)).$
fong-spivak-invitation_EQ0140	221	■ 1.000	$\operatorname{Circ}(V) := \{(V, A, s, t, \ell) \mid \text{where } s, t: A \rightarrow V, \ell: E \rightarrow C\} \text{ .}$	$\operatorname{Circ}(V) := \{(V, A, s, t, \ell) \mid \text{ where } s, t: A \rightarrow V, \ell: E \rightarrow C\}.$	$\operatorname{Circ}(V) := \{(V, A, s, t, \ell) \mid \text{ where } s, t: A \rightarrow V, \ell: E \rightarrow C\}.$
fong-spivak-invitation_EQ0147	230	■ 1.000	$F: \mathcal{O}\left(t_1, \ldots, t_n\right) ; t \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right) \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right) \text{ ; } f(t) \text{ .}$	$F: \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{P}(f(t_1), \dots, f(t_n); f(t))$	$F: \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{P}(f(t_1), \dots, f(t_n); f(t))$
fong-spivak-invitation_EQ0154 (7.6)	241	■ 1.000	$X = \{y \in Y \mid p(y) = \text{true}\} \text{ .}$	$X = \{y \in Y \mid p(y) = \text{true}\}.$	$X = \{y \in Y \mid p(y) = \text{true}\}.$
fong-spivak-invitation_EQ0158	249	■ 1.000	$\left.s_i\right _{U_i \cap U_j} = \left.s_j\right _{U_i \cap U_j} \text{ .}$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$
fong-spivak-invitation_EQ0162 (7.14)	255	■ 1.000	$\begin{aligned} & \Omega(U) \rightarrow \Omega(V), \\ & U' \mapsto U' \cap V. \end{aligned}$	$\begin{aligned} & \Omega(U) \rightarrow \Omega(V), \\ & U' \mapsto U' \cap V. \end{aligned}$	$\begin{aligned} & \Omega(U) \rightarrow \Omega(V), \\ & U' \mapsto U' \cap V. \end{aligned}$
fong-spivak-invitation_EQ0163	256	■ 1.000	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j \text{ .}$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j.$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j.$
fong-spivak-invitation_EQ0169	261	■ 1.000	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$
fong-spivak-invitation_EQ0173	266	■ 1.000	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S\left(o_{[a-\epsilon, b+\epsilon]}\right)$	$S(o_{[a, b]}) \cong \lim_{\epsilon > 0} S(o_{[a - \epsilon, b + \epsilon]})$	$S(o_{[a, b]}) \cong \lim_{\epsilon > 0} S(o_{[a - \epsilon, b + \epsilon]}).$
fong-spivak-invitation_EQ0175	271	■ 1.000	$\varnothing, \quad \{1\}, \quad \{2\}, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\}, \quad \{1, 2, 3\} \text{ .}$	$\varnothing, \quad \{1\}, \quad \{2\}, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\}, \quad \{1, 2, 3\}.$	$\varnothing, \quad \{1\}, \quad \{2\}, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\}, \quad \{1, 2, 3\}.$
fong-spivak-invitation_EQ0176	271	■ 1.000	$(h, 1), \quad \quad (h, 2), \quad \quad (h, 3), \quad \quad (1, 1), \quad \quad (1, 2), \quad \quad (1, 3) \text{ .}$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$
fong-spivak-invitation_EQ0177	271	■ 1.000	$(h, 1), \quad \quad (1, 1), \quad \quad (1, 2), \quad \quad (2, 2), \quad \quad (3, 2) \text{ .}$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$
fong-spivak-invitation_EQ0178	271	■ 1.000	$h, \quad \quad 1, \quad \quad 2, \quad \quad 3 \text{ .}$	$h, \quad 1, \quad 2, \quad 3.$	$h, \quad 1, \quad 2, \quad 3.$

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fong-spivak-invitation-EQ0179	272	■ 1.000	$\begin{array}{lllll} (11,11), & \& (11,12), & \& (12,11), & \& \\ (12,12), & \& (13,13), & \& (21,21), & \& (22,22), & \& (22,23), & \& \\ (23,22), & \& (23,23) . \end{array}$	$(11,11), (11,12), (12,11), (12,12), (13,13), (21,21), (22,22), (22,23), (23,22), (23,23).$	$(11,11), (11,12), (12,11), (12,12), (13,13), (21,21), (22,22), (22,23), (23,22), (23,23).$
fong-spivak-invitation-EQ0182	273	■ 1.000	$(\bullet)(\circ)(*), \quad \quad \quad (\bullet\circ)(*), \quad \quad \quad (\bullet*)(\circ), \quad \quad \quad (\bullet)(\circ*), \quad \quad \quad (\bullet\circ*).$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*).$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*).$
fong-spivak-invitation-EQ0184	280	■ 1.000	$g(q) \leq g^{\{\prime\}}(f(g(q))) \leq g^{\{\prime\}}(q) .$	$g(q) \leq g'(f(g(q))) \leq g'(q).$	$g(q) \leq g'(f(g(q))) \leq g'(q).$
fong-spivak-invitation-EQ0188	288	■ 1.000	$A=(A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C .$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$
fong-spivak-invitation-EQ0190	288	■ 1.000	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$
fong-spivak-invitation-EQ0194	289	■ 1.000	$\begin{gathered} M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right), \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right), \\ M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right). \end{gathered}$	$M_X = \left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right), \quad M_X^2 = \left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right),$ $M_X^3 = M_X^4 = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right).$	$M_X = \left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right), \quad M_X^2 = \left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right),$ $M_X^3 = M_X^4 = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right).$
fong-spivak-invitation-EQ0196	290	■ 1.000	$(p \ ; \ q):=\left(v, a_{\{1\}}, \ldots, a_{\{m\}}, b_{\{1\}}, \ldots, b_{\{n\}}\right) .$	$(p;q):=(v,a_1,\ldots,a_m,b_1,\ldots,b_n).$	$(p\mathbin{\circ}q):=(v,a_1,\ldots,a_m,b_1,\ldots,b_n).$
fong-spivak-invitation-EQ0197	290	■ 1.000	$\left(v, a_{\{1\}}, \ldots, a_{\{m\}}, b_{\{1\}}, \ldots, b_{\{n\}}, c_{\{1\}}, \ldots, c_{\{o\}}\right) .$	$(v,a_1,\ldots,a_m,b_1,\ldots,b_n,c_1,\ldots,c_o).$	$(v,a_1,\ldots,a_m,b_1,\ldots,b_n,c_1,\ldots,c_o).$
fong-spivak-invitation-EQ0203	292	■ 1.000	$\begin{array}{lll} (1,1) & \& (1,2) & \& (1,3) \\ (2,1) & \& (2,2) & \& (2,3) \\ (3,1) & \& (3,2) & \& (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$
fong-spivak-invitation-EQ0204	292	■ 1.000	$f^{\{-1\}}(1)=b, \quad \quad \quad f^{\{-1\}}(2)=a, \quad \quad \quad f^{\{-1\}}(3)=c .$	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c.$	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c.$
fong-spivak-invitation-EQ0205	293	■ 1.000	$\begin{gathered} A^{\{\prime\}} \mapsto A, \quad \quad \quad f^{\{\prime\}} \mapsto f, \quad \quad \quad g^{\{\prime\}} \mapsto g, \quad \quad \quad h^{\{\prime\}} \mapsto h, \quad \quad \quad i^{\{\prime\}} \mapsto i, \quad \quad \quad D^{\{\prime\}} \mapsto D . \end{gathered}$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f';h' \mapsto f;h, \quad g';i' \mapsto f;h, \quad B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f' \mathbin{\circ} h' \mapsto f \mathbin{\circ} h, \quad g' \mathbin{\circ} i' \mapsto f \mathbin{\circ} h, \quad B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$
fong-spivak-invitation-EQ0208	293	■ 1.000	$((F \ ; \ G) \ ; \ H)(c)=H((F \ ; \ G)(c))=H(G(F(c)))=(G \ ; \ H)(F(c))=(F \ ; \ (G \ ; \ H))(c) .$	$((F;G);H)(c) = H((F;G)(c)) = H(G(F(c))) = (G;H)(F(c)) = (F;(G;H))(c).$	$((F \mathbin{\circ} G) \mathbin{\circ} H)(c) = H((F \mathbin{\circ} G)(c)) = H(G(F(c))) = (G \mathbin{\circ} H)(F(c)) = (F \mathbin{\circ} (G \mathbin{\circ} H))(c).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_EQ0211	297	■ 1.000	$\left(f_{\{1\}} ; f_{\{2\}}\right)^B(g)=g ; \left(f_{\{1\}} ; f_{\{2\}}\right)=\left(g ; f_{\{1\}}\right) ; f_{\{2\}}=\left(f_{\{1\}}^B ; f_{\{2\}}^B\right) ; f_{\{2\}}^B\right)(g)$	$(f_1;f_2)^B(g)=g;(f_1;f_2)=(g;f_1);f_2=(f_1^B;f_2^B)(g)$	$(f_1\circ f_2)^B(g)=g\circ(f_1\circ f_2)=(g\circ f_1)\circ f_2=(f_1^B\circ f_2^B)(g)$
fong-spivak-invitation_EQ0218	300	■ 1.000	$\begin{array}{c ccc} \Phi & x & y & z \\ \hline A & 17 & 20 & 20 \\ B & 11 & 14 & 14 \\ C & 14 & 17 & 17 \\ D & 12 & 9 & 15 \end{array}$	$\begin{array}{c ccc} \Phi & x & y & z \\ \hline A & 17 & 20 & 20 \\ B & 11 & 14 & 14 \\ C & 14 & 17 & 17 \\ D & 12 & 9 & 15 \end{array}$	$\begin{array}{c ccc} \Phi & x & y & z \\ \hline A & 17 & 20 & 20 \\ B & 11 & 14 & 14 \\ C & 14 & 17 & 17 \\ D & 12 & 9 & 15 \end{array}$
fong-spivak-invitation_EQ0229	307	■ 1.000	$(f+g)(i):=\begin{cases} f(i) & \text{if } i\leq m, \\ g(i-m)+m^{\prime} & \text{if } m+1\leq i. \end{cases}$	$(f+g)(i):=\begin{cases} f(i) & \text{if } i\leq m, \\ g(i-m)+m' & \text{if } m+1\leq i. \end{cases}$	$(f+g)(i):=\begin{cases} f(i) & \text{if } i\leq m, \\ g(i-m)+m' & \text{if } m+1\leq i. \end{cases}$
fong-spivak-invitation_EQ0230	310	■ 1.000	$a^{\{0\}}, a^{\{1\}}, a^{\{2\}}, a^{\{3\}}, \ldots, a^{\{2019\}}, \ldots$	$a^0,a^1,a^2,a^3,\ldots,a^{2019},\ldots$	$a^0,a^1,a^2,a^3,\ldots,a^{2019},\ldots$
fong-spivak-invitation_EQ0233	311	■ 1.000	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$
fong-spivak-invitation_EQ0234	311	■ 1.000	$A*B=\left(\begin{array}{ll} 0&1\\ 0&0 \end{array}\right)*\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)\neq\left(\begin{array}{ll} 0&1\\ 1&0 \end{array}\right)*\left(\begin{array}{ll} 0&1\\ 0&0 \end{array}\right)=B*A$	$A*B=\left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right)*\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right)\neq\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right)*\left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right)=B*A.$	$A*B=\left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right)*\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right)\neq\left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right)*\left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right)=B*A.$
fong-spivak-invitation_EQ0236	312	■ 1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-invitation_EQ0237	312	■ 1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-invitation_EQ0238	313	■ 1.000	$g_{\{1\}}:=c_{\{n\}}+\cdots+c_{\{n\}}:m\rightarrow(m\times n),$	$g_1:=c_n+\cdots+c_n:m\rightarrow(m\times n),$	$g_1:=c_n+\cdots+c_n:m\rightarrow(m\times n),$
fong-spivak-invitation_EQ0240	313	■ 1.000	$\begin{aligned} g_{\{2\}}:=& \ s_{\{M(1,1)\}}+\cdots+s_{\{M(1, n)\}}\ \& \ +s_{\{M(2,1)\}}+\cdots+s_{\{M(2, n)\}}\ \& \ +\cdots \\ & \ +s_{\{M(m, 1)\}}+\cdots+s_{\{M(m, n)\}}:(m\times n) \\ & \rightarrow(m\times n), \end{aligned}$	$\begin{aligned} g_2:=& s_{M(1,1)}+\cdots+s_{M(1,n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2,n)} \\ & +\cdots \\ & +s_{M(m,1)}+\cdots+s_{M(m,n)}:(m\times n)\rightarrow(m\times n), \end{aligned}$	$\begin{aligned} g_2:=& \ s_{M(1,1)}+\cdots+s_{M(1,n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2,n)} \\ & +\cdots \\ & +s_{M(m,1)}+\cdots+s_{M(m,n)}:(m\times n)\rightarrow(m\times n), \end{aligned}$
fong-spivak-invitation_EQ0241	313	■ 1.000	$g_{\{4\}}:=a_{\{m\}}+\cdots+a_{\{m\}}:(m\times n)\rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m\times n)\rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m\times n)\rightarrow n,$
fong-spivak-invitation_EQ0243	315	■ 1.000	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$
fong-spivak-invitation_EQ0244	315	■ 1.000	$U(\mu)\left(\left(a_{\{1\}}, \ldots, a_{\{m\}}\right), \left(b_{\{1\}}, \ldots, b_{\{m\}}\right)\right):=\left(a_{\{1\}}+b_{\{1\}}, \ldots, a_{\{m\}}+b_{\{m\}}\right)$	$U(\mu)\left((a_1,\ldots,a_m),(b_1,\ldots,b_m)\right):=(a_1+b_1,\ldots,a_m+b_m).$	$U(\mu)\left((a_1,\ldots,a_m),(b_1,\ldots,b_m)\right):=(a_1+b_1,\ldots,a_m+b_m).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																											
fong-spivak-invitation-EQ0246	316	1.000	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$																											
fong-spivak-invitation-EQ0249	316	1.000	$\begin{aligned} \mathrm{B}(g^{\mathrm{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g^{\mathrm{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g^{\mathrm{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$																											
fong-spivak-invitation-EQ0251	316	1.000	$\mathrm{B}(\mathrm{left}(\mathrm{left}(g^{\mathrm{op}})\mathrm{right}) ; h\mathrm{right}) = \mathrm{left}(\{(S(g)(x), S(h)(x)) \mid x \in R^m\}\mathrm{right}) .$	$\mathrm{B}((g^{\mathrm{op}}); h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\} .$	$\mathrm{B}((g^{\mathrm{op}}) \circ h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\} .$																											
fong-spivak-invitation-EQ0252	318	1.000	$f(m) = f(\underbrace{1 + 1 + \cdots + 1}_m \text{ summands}) = \underbrace{f(1) + f(1) + \cdots + f(1)}_m \text{ summands} = \underbrace{1_R + 1_R + \cdots + 1_R}_m \text{ summands} .$	$f(m) = f(\underbrace{1 + 1 + \cdots + 1}_m \text{ summands}) = \underbrace{f(1) + f(1) + \cdots + f(1)}_m \text{ summands} = \underbrace{1_R + 1_R + \cdots + 1_R}_m \text{ summands} .$	$f(m) = f(\underbrace{1 + 1 + \cdots + 1}_m \text{ summands}) = \underbrace{f(1) + f(1) + \cdots + f(1)}_m \text{ summands} = \underbrace{1_R + 1_R + \cdots + 1_R}_m \text{ summands} .$																											
fong-spivak-invitation-EQ0254	318	1.000	$\begin{aligned} [f, g]: A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$	$\begin{aligned} [f, g]: A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$	$\begin{aligned} [f, g]: A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$																											
fong-spivak-invitation-EQ0261	338	1.000	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$																											
fong-spivak-invitation-EQ0262	338	1.000	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$																											
fong-spivak-invitation-EQ0064	114	—	—	—	$G := \begin{array}{c} 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \end{array} \quad H := \begin{array}{c} 4 \xrightarrow[c]{d} 5 \curvearrowright e \end{array}$																											
fong-spivak-invitation-EQ0065	114	—	—	—	<table><tr><td rowspan="2">$G :=$</td><td>Arrow</td><td>source</td><td>target</td><td>Vertex</td></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td rowspan="3">$H :=$</td><td>b</td><td>2</td><td>3</td><td>2</td></tr><tr><td>c</td><td>4</td><td>5</td><td>3</td></tr><tr><td>d</td><td>4</td><td>5</td><td>4</td></tr><tr><td rowspan="2"></td><td>e</td><td>5</td><td>5</td><td>5</td></tr></table>	$G :=$	Arrow	source	target	Vertex	a	1	2	1	$H :=$	b	2	3	2	c	4	5	3	d	4	5	4		e	5	5	5
$G :=$	Arrow	source	target	Vertex																												
	a	1	2	1																												
$H :=$	b	2	3	2																												
	c	4	5	3																												
	d	4	5	4																												
	e	5	5	5																												